
UNDERGROUND COAL MINING

Course Code : 332307

Programme Name/s : Mining & Mine Surveying/ Mining Engineering**Programme Code : MS/ MZ****Year : Second****Course Title : UNDERGROUND COAL MINING****Course Code : 332307****I. RATIONALE**

It is important that the mining diploma students should have knowledge about the common methods of working coal with special reference to Indian Coal Mining. This subject is introduced to understand Methods of Mining of Coal e.g. Board and Pillar working , development and depillaring, Long wall methods both advancing and retreating, special methods for working under special difficult situation and of contiguous seams and thick seam Working. Since new working methods like High wall Mining, Punch Long wall Mining and blasting Gallery method are likely to be practiced in future , these topics have been included this course.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Aim of this course is to help the student to attain the following industry identified outcomes . Select suitable Method of working for underground Coal Mining and Design suitable layouts for Board and Pillar and Long wall Mining.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Use appropriate method of working under given geological condition to prepare suitable layout.
- CO2 - Select type of depillaring method for existing roof conditions of the underground coal mines.
- CO3 - Choose the appropriate method of thick seam working and prepare a working plan.
- CO4 - Design long wall working layout under given geological conditions.

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- CO5 - Identify the effect of subsidence due to underground working to control .

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
															Practical						
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
Max	Max	Max	Min	Max	Min	Max	Min	Max	Min												
332307	UNDERGROUND COAL MINING	UCM	DSC	3	-	2	-	5	5	3	30	70	100	40	25	10	25#	10	-	-	150

Total IKS Hrs for Sem. : 1 Hrs

Abbreviations: CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @\$ Internal Online Examination

Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 30 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

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Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Classify the given types of method of working. TLO 1.2 Select the suitable method of working. TLO 1.3 Calculate Manpower and OMS under given condition. TLO 1.4 Compare panel System of working on the basis of given criteria.	Unit - I Board and Pillar Method (Development) 1.1 Classification of methods of working ,stripping ratio ,geological parameters , technical parameters, history of Indian Coal Mining (IKS). 1.2 Conditions for selection of method of working, Board and pillar applicability advantages and disadvantages, different types development (Strike and Dip development), their advantageous and disadvantages. 1.3 Calculation of percentage of extraction, Design of panel, Different layout classification ,SDL chain conveyor layout ,Continuous miners ,LHD and chain conveyor layout ,Manpower calculation and OMS. 1.4 Open and closed panel system. advantageous and disadvantages ,isolation stopping (Ordinary and explosion proof).	Model Demonstration Video Demonstrations Lecture Using Chalk-Board Presentations

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Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Execute Preparatory arrangements to be done before depillaring. TLO 2.2 Explain types of pillar extraction methods. TLO 2.3 Describe procedure of Goaf Treatment.	Unit - II Board and Pillar Method (Depillaring) 2.1 Preparatory arrangements before depillaring, permission from different regulatory authorities, check and joint survey, updatation of plans and sections, construction of isolation and preparatory isolation stopping, Line of extraction and numbering of pillars, Systematic support rules. 2.2 Different types of pillar extraction methods depending upon the roof condition, spitting and slicing , slicing, extraction , partial extraction, formation of Goaf, withdrawal of support. 2.3 Caving, definition of local fall, main fall, cavability index, dangers associated with caving, precautioid air blast, Hydraulic Stowing, definition of hydraulic profile, (H/L Ratio), Surface and underground arrangements to be done for carrying hydraulic stowing.	Model Demonstration Video Demonstrations Lecture Using Chalk-Board Site/Industry Visit

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Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Describe the procedure of working of coal seam in Board and Pillar Method. TLO 3.2 Follow safety precautions while working near restricted areas. TLO 3.3 Follow safety precautions while Working near fire area. TLO 3.4 Follow safety precautions while working below waterlogged area. TLO 3.5 Follow safety precautions while working below depillared goaf .	Unit - III Thick seam and Contiguous Seam Working 3.1 Thick seam and Contiguous Seam Working, precautions to be taken while working contiguous seams. Layout and case study. 3.2 Precautions while working near restricted areas as per CMR 2017. 3.3 Working near fire area as per CMR 2017. 3.4 Working below waterlogged area as per CMR 2017. 3.5 Working below depillared goaf as per CMR 2017.	Model Demonstration Video Demonstrations Presentations Site/Industry Visit

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Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Design the longwall Panel under given geological condition. TLO 4.2 Describe different types of longwall mining methods.	Unit - IV Longwall Mining 4.1 Applicability of longwall mining, Design of long wall panel, Factors affecting length of longwall face, Barrier width, gate road length. 4.2 Different types of Longwall Mining, Longwall advancing, Longwall retreating ,Cyclic longwall ,Non cyclic long wall, single unit face and double unit face, different cutting machines used Single ended ranging drum shearer (SERD) and Double ended ranging drum shearer (DERD) Layout of DERD manpower calculation.	Video Demonstrations Presentations Lecture Using Chalk-Board Site/Industry Visit
5	TLO 5.1 Describe the theories of subsidence. TLO 5.2 Identify critical factors which causes subsidence. TLO 5.3 Elaborate precautions to be taken for reducing subsidence. TLO 5.4 Explain the techniques of measurement of subsidence.	Unit - V Subsidence due to Underground Mining 5.1 Different Theories of subsidence, Normal Theory, Vertical Theory, in-between Normal and vertical, Dome theory, Different definition related to subsidence, different types of failure. 5.2 Different factors affecting subsidence, Depth Thickness, nature of strata, method of working, relation of subsidence with time. 5.3 Precautions to be taken on surface and underground to reduce subsidence Concept of Harmless Depth. 5.4 Measurement of subsidence and preservation of record, various techniques used, Measurement of Subsidence using Survey.	Model Demonstration Video Demonstrations Case Study Lecture Using Chalk-Board

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.**MSBTE Approval Dt. 01/10/2024****Semester - 2, K Scheme**

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Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Workout the percentage of extraction during development.	1	Calculation of the percentage of extraction during development under given geological conditions of mining.	2	CO1
LLO 2.1 Prepare layout of Development by Side Discharge Loader (SDL)/Load Haul Dumper (LHD) with chain conveyor.	2	Drawing layout of development by SDL/LHD with chain Conveyor. and Calculate Manpower and Output per Men per Shift (OMS) by assuming your own Conditions.	4	CO1
LLO 3.1 Prepare a layout of Development by Continuous Miner with shuttle cars.	3	Drawing layout of development by Continuous Miner with shuttle cars and calculate Manpower and OMS by assuming your own Conditions.	4	CO1
LLO 4.1 Estimate the number of pillars in closed panel.	4	Calculation of number of pillars in closed panel under given conditions.	2	CO1
LLO 5.1 Draw Ordinary Isolation stopping and Explosion proof isolation stopping.	5	Creation of Ordinary Isolation stopping and Explosion proof isolation stopping.	2	CO1
LLO 6.1 Evolve suitable arrangements to be made before starting depillaring.	6	Preparation of panel for depillaring. Suggest the suitable arrangements to be done before starting depillaring.	2	CO2
LLO 7.1 Plan the line of extraction.	7	Design of line of extraction under given roof conditions and other factors.	2	CO2
LLO 8.1 Suggest suitable Strata Control and Monitor Plan.	8	Creation of Strata Control and Monitor Plan under given conditions.	2	CO2
LLO 9.1 Prepare the layout of different types of extraction of coal pillars.	9	Drawing of layout of different types of extraction of coal pillars under given conditions.	2	CO2

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Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 10.1 Prepare layout of Sand Stowing.	10	Drawing of layout of Sand Stowing in underground coal mines.	2	CO2
LLO 11.1 Identify the dangers associated with caving method. LLO 11.2 Suggest suitable remedial measures.	11	Identification of dangers associated with caving method and suggest suitable remedial measures.	2	CO2
LLO 12.1 Draw layout of development of contiguous working. LLO 12.2 Prepare layout of depillaring of contiguous working.	12	Drawing layout of development of contiguous working and Drawing layout of depillaring of contiguous working.	4	CO3
LLO 13.1 Develop the working plan for working near restricted areas.	13	Preparation of working plan for working near restricted areas.	2	CO3
LLO 14.1 Execute the working plan for Working near fire area.	14	Preparation of working plan for Working near fire area.	2	CO3
LLO 15.1 Ensure the working plan for Working below waterlogged area.	15	Preparation of working plan for Working below waterlogged area.	2	CO3
LLO 16.1 Ensure the working plan for Working below depillared goaf.	16	Development of working plan for Working below depillared goaf.	2	CO3
LLO 17.1 Sketch a layout of Longwall Advancing. LLO 17.2 Sketch a layout of Longwall retreating.	17	Drawing a layout of Longwall Advancing and layout of Longwall retreating.	4	CO4

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Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 18.1 Sketch a layout of Longwall Single unit Face. LLO 18.2 Sketch a layout of Longwall Double unit Face.	18	Draw a layout of Longwall Single unit Face and. Draw a layout of Longwall Double unit Face.	4	CO4
LLO 19.1 Prepare a layout of Non cyclic Longwall Mining using DERD. LLO 19.2 Calculate manpower and OMS for prepared layout.	19	Drawing a layout of Non cyclic Longwall Mining using DERD and Calculation of manpower and OMS.	4	CO4
LLO 20.1 Sketch Angle of Draw , Limit Line of subsidence basin. LLO 20.2 Sketch Normal plane ,vertical plane of subsidence basin.	20	Interpretion of different theories of Subsidence. and Drawing a Angle of Draw , Limit line , Normal plane ,vertical plane.	4	CO5
LLO 21.1 Sketch a vertical, angle of break. LLO 21.2 Sketch mode of failure , subsidence basin.	21	Interpretion of different terminologies used in Subsidence. and Drawing a vertical, angle of break. and drawing mode of failure of subsidence basin.	2	CO5
LLO 22.1 Identify different types of methods used for measurement of subsidence.	22	Selection of different types of methods used for measurement of subsidence.	2	CO5
Note : Out of above suggestive LLOs - • '*' Marked Practicals (LLOs) Are mandatory. • Minimum 80% of above list of lab experiment are to be performed. • Judicial mix of LLOs are to be performed to achieve desired outcomes.				

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VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)**Micro project**

- Visit any underground coal mine and Draw the layout of Depillaring and calculate OMS.
- Prepare PPT and brief report of histroy of Mining activities in ancient India.
- Visit any underground coal mine and Draw the layout of development and calculate OMS.

Assignment

- Collect the data of nearby mines where DERD/SERD is using for production.
 - Collect the information of Subsidence (Case Study).
 - Collect the information of nearby mines working with long wall mining methods, machineries, supports, production etc
 - Articulate case study of Thick Seam Working in any underground coal mine.
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- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicious mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Model of Coal Pillar showing extraction of gallery	1
2	Model showing Hydraulic sand stowing in depillaring area	10
3	Model showing caving method with safety	11
4	Model Showing working of Contiguous seam	12
5	Model showing plan of working below depillared goaf area.	16
6	Model showing Longwall Advancing method.	17
7	Model showing Longwall Retreating method.	17
8	Model showing working of Single unit and double unit Long wall method.	18
9	Smart Classroom fitted with computer and DLP	19,20,21,22
10	Model of layout of Development district by using SDL/LHD	2

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Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
11	Model of layout of Development district by using Continuous Miner	3
12	Model showing closed panel system of working	4
13	Model of Ordinary and Explosion proof Isolation Stopping	5
14	Model of depillaring district.	6
15	Model showing different Line of Extraction methods.	7
16	Model showing Application of Strata Contrl and Monitoring Plan in depillaring district	8
17	Model showing Stooking, splitting & Slicing of pillar Extraction.	9

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Board and Pillar Method (Development)	CO1	20	2	8	6	16
2	II	Board and Pillar Method (Depillaring)	CO2	20	2	8	6	16
3	III	Thick seam and Contiguous Seam Working	CO3	12	2	4	4	10
4	IV	Longwall Mining	CO4	24	2	6	12	20
5	V	Subsidence due to Underground Mining	CO5	14	2	2	4	8
Grand Total				90	10	28	32	70

X. ASSESSMENT METHODOLOGIES/TOOLS**Formative assessment (Assessment for Learning)**

- First Progressive Test

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- Second Progressive test
- Lab Performance
- Journal Submission

Summative Assessment (Assessment of Learning)

- End Year Examination
- Viva -voce

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	3	2	2	1			2			
CO2	3	2	3	1			1			
CO3	3	2	1	1			3			
CO4	3	2	1	1			2			
CO5	3	2	1	1			1			

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Legends :- High:03, Medium:02,Low:01, No Mapping: -
 *PSOs are to be formulated at institute level

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	R. D. Singh	Principles & Practices of Modern Coal Mining	New Age International (P) Limited, Publishers, New Delhi ISBN: 81-224-0974-1
2	Heartman L. Howard	Introductory Mining Engineering	John Wiley & Sons, New York, ISBN: 0-471-62804-2
3	G.K. Pradhan	Explosive and Blasting Technology	Lovely Prakashan, Dhanbad
4	L.C. Kaku	Mining Digest	Lovely Prakashan, Dhanbad
5	CIMFR	Ground Control in Board and Pillar Mines	Lovely Prakashan, Dhanbad
6	Dr B.P. Verma	Roof Bolting	Lovely Prakashan, Dhanbad
7	D.J.Deshmukh	Elements of Mining Technology	Lovely Prakashan, Dhanbad

XIII. LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://www.youtube.com/watch?v=oDduxw9d6lw	videos regarding mining
2	https://discoveryjournals.org/discoveryengineering/index.htm	videos regarding mining
3	https://www.slideshare.net/pramodgpramod/coal-mining-methods-74267944?from_search=0	coal mining methods
4	https://youtu.be/wjNOG36i0Aw?si=Y9TXqLiEgApcWHKF	subsidence

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Sr.No	Link / Portal	Description
5	https://www.slideshare.net/anilitbhu/longwall-top-coal-caving-g-method?from__search=7	longwall top coal caving method
6	https://www.slideshare.net/shanz2u/longwall-mining-14602171?from__search=0	longwall mining

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students

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